

ActInf Textbook Group ~ Cohort 1 ~ Meeting 16

(Chapter 6, part 2)

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okay hello it's September 16th we're in cohort one of the active textbook group it's uh meeting 15. or I guess uh yes no 16. on um let me just update the title meaning 16 we're in our second discussion on chapter six so are there any overall comments that people would like to address about chapter six or any other questions or points to add or just areas of chapter six that they found to be interesting distinction between the interceptive X receptive uh modalities which have been mentioned in box 6.1 and first I think Kristen with my Michael Allen if I'm not mistaken uh has published a recent paper just today uh exactly on that topic in which they've somehow tried to provide a computational model for the first time for the unification of the intercepted the next receptive modalities so since this chapter is also that the modeling of those agents I think this paper from in the body's eye the computational anatomy of interceptive inference might be relevant for anyone who wants to model these modalities for the aid of active inference framework thank you very nice thanks for sharing it um let's look at box six one as people see it what distinguishes exteroceptive proprioceptive and interoceptive modalities for example why are proprioceptive and interceptive separated and because uh I couldn't hear you I don't know if it's on my side or oh connection was unstable from your side sorry let me try something I'll be gone for a second am I back okay so yeah what uh yeah Brock I do need the starlink why are proprioception and terraception separated and why are these three categories modeled as different why have they been framed as different biological and neurological mechanisms and neural mechanisms that somehow can control those different kind of perceptions because uh for the extraceptive perception the organism would necessarily meet would necessarily need a kind of sensory I mean the sensation it gets from the environment is just um on the level of its uh of the imminent stimuli that it gets from the external environment but for the interceptive uh modality uh the sensation is just the internal uh Sensations and so on and for the proprioceptive it's related to motor control uh feedback loop so I guess because of their different mechanisms it might be helpful to talk in different

terms about each of them but I'm not totally sure about that I agree I think one aspect even in a pre-blanket or non-blanket modality is like they're about external things versus about internal things and then between proprioception and interception you're absolutely right that they have different neural mechanisms and physiological mechanisms um with some blurry space perhaps for example like interception about blood pressure is somewhere in between like a muscular reflex and uh in organ reflex but the tissues being addressed are different let's look at this paper again we'll we'll return to the book but we're going to be thinking today and then especially going forward with like how do you identify the let's go to this question which we we just just to look at it which system are we modeling what is the appropriate form for the generative model how do we set up a generative model how do we set up a generative process so let's see if we can determine some of those aspects in this paper #figure four three it is the very same okay this is just a general pointing to a partially observable Markov decision process with a slight um plus computational biology graphical veneer to match the color scheme probably not but but it kind of looks like it here is where we see this schematic of the form of the generative model variables are used to reflect that those variables can be fit from empirical data or just a priori set and simulate it but this is the form of the model these are the shapes of the matrices and tensors that go from that are going to address what their model is about um does anybody want to like describe what they see in this what do you think they were trying to explain and what do you think they were trying to model what is their system of Interest based upon what you see here uh actually one interesting term that came up with I mean used in this paper is uh cardiac active inference the term that they've been they've used in this paper to describe uh the kind of interrelationship between the heart rhythm and uh the I mean the interception so this generative model uh I believe is uh just about modeling that relationship between the heart rhythm and the interception so uh we can see as usual the B matrices c and a matrices that we've seen before and but in this case uh they've applied uh to the uh variation in the heart rhythm and how it relates to the interception and so yes so again using the variables that we know from figure four three in the book and their figure one here the hidden states that are not directly observed are on the interceptive side relating to the state of the heart on the exteriorceptive side whether there is a flower or a spider the outcomes that are observed are the visual exteroception of a flower or a spider and the heart rate probably a low and a high heart rate the piece that links these two is the sensory attenuation between the high heart rate states and the mapping of exteroception in the B Matrix you can just see that even the the allocation of the symbols is different like when you're aroused when you're in this

this is like heart open and then heart contract so when you're in the aroused situation B is how the hidden States change their time you're always going to go from here to here whereas in the relaxed situation sometimes you go to the from this state to this one other times you're relaxed and so you actually you dwell within that state so here it's like pump rest pump rest pump rest it's 50 slower just it depends on the parameters but it's 50 slower because sometimes you're going to go from one but it looks like as it's here that it's actually like or it's twice as slow rest rest beat rest rest beat so these are the two interceptive states one has a preference for having relaxed Hearts when flowers are around and having contracted heart when spiders the a matrix maps on these two sensory between hidden States and outcome States just like any a matrix would do in any pomdp and then there's also a transition Matrix of the generative process so it's a very simple generative process where either flowers or flowers or spiders or spiders as they would say you can use standard routines to simulate these techniques I hope that that the code are available you can plot parameter sweeps as you explore the beta and the alpha yeah so here's like the beta parameter and Alpha parameter so you can do parameter sweeps with this grid search or other approaches and then do statistics on those outputs is this quote active inference well it's the output of a parameter sweep on an active inference model there's various other kinds of methods so not to spend too too much time on the paper but does anyone have any thoughts on like case or how they see it applying to chapter six's content I think it speaks to a um fulfillment cycle in active inference research oh this work example helps a lot I mean if anyone wants to uh especially because uh describing this kind of cardiac active inference as a kind of system that has a definite Mark of blanket uh it was a really interesting question at the first place and uh I think they've done uh really um great job obviously in that regard and thinking really creatively and out of the box uh about how we can describe that whole system uh with Native using markup blankets to describe its internal and external environment and so on it also um relates to this earlier question so how does the figure four three paradigm discrete and continuous State spaces with all the attending complexities do you hear me Ellie okay yeah okay okay how does this representation connect to um can you hear Daniel because yeah it's a story for another day some of the incredible the truly incredible internet I'm gonna try one thing so I think that or Holly I'll try something foreign here's what's interesting with the Wi-Fi and then we'll continue because it's not it's not a Wi-Fi learning group but with the hot spot there are a set of people who I cannot interact with but with Wi-Fi I am able to interact with those people just a note we see the pomdp representation now what is not in this paper here is a graphical layout of the pomdp okay a figure six one or indeed a figure one one

representation so why is that uh I'm not sure I've understood your question quite clearly but uh could you please elaborate on that a little bit yes yes sure so when we look at papers like Vassal at all communication is active inference social and active active inference loops loops loops you know Falcon at all remote teams loops loops loops Markov blanket interface Loops entities things yet in the application what is specified is not the loopy situation what needs to be specified is chapter six not chapter one so I find that very interesting because so much of the focus attention discourse has been around abstractions around Markov blankets and their interpretations from various stances and about boundaries physical boundaries and conceptual boundaries but it's a non-controversy when it's written this way and there are Markov blankets between many of these variables because it's a Bayesian graph that's not fully connected so every Bayesian graph that's not fully connected has Markov blankets trivially but this paper doesn't need to describe what is this is enough to say what's inside and outside the organismal Markov blanket if you want to even think about it is like here in this line kind of not not on the base graph but it's between The Spider and the flower outside and the vision of The Spider and the flower but that's like taken care of with the a matrix of the exteroceptive modality which is I think why this recipe does not focus on which thing are we modeling but rather which system are we modeling and yes we're going to pursue the generative model before the generative process even though they may have symmetry or they may be very similar but I just find it very interesting open to hearing what people think like why the cybernetic Loop action perception Loop is used so much in contrast with the form of the model that's actually implemented which is a it's a different topology in a different discussion I know that they're integrated and compatible uh just one comment on that is you see at least to my understanding uh Markov blanket is a kind of uh the kind of think modeling tool or a kind of uh a way of thinking about the basically the ontology of I mean the Anthology in the sense of philosophy it's philosophical use the ontology of the system that we're about to model so uh it's not necessarily uh the relationships among the different components of the Markov blanket is there's something not necessarily physical or tangible relationships in that system so it might be uh kind of just informational as as we've just we saw in I think chapter five if I'm not mistaken uh the information passing among the different components and so on so uh at least I think the reason for not using uh the schematic model of Markov I mean the cybernetic model of a Markov blanket instead of uh using just the schematic form uh probably it's because of this uh kind of uh non-let's say non-physical or non-tangible relationship among the different components by non-tangible and non-physical I mean something that can't be explained in

terms of its uh ontological existence if I don't know if I would I could express myself in that regard or not but thank you ali um Brock and anyone else um I was just going to I don't know um I I feel like personally I'm just kind of at my um the affordances that I have to get pragmatic and epistemic value here without like doing some modeling of some kind yeah I don't know that's just good thanks I mean this is the the second discussion on chapter six and then this is setting us up to actually pursue a model so we should be providing these things yes I'm just yeah I'm just saying it it's definitely ripe for that I I mean yes uh Rohan uh yeah so uh actually just to get some clarification you're asking why the form of the model is as mentioned in Japanese given in chapter six which is chapter one yes I mean they're they're compatible topologies that actually are describing an overlapping set of notes I was just calling attention to the the um amount of discourse and focus on the potentially more familiar more schematic action perception Loop and the way that that draws so much attention to the organism environment interface whereas when it comes to the empirical modeling that interface never had to be specified as like a first class type it just was included by the by we just said there are spiders and flowers out there hidden States s and then there's observations of and and there I there's no error there's no errors there would be no error in the generative process itself the model might have better right so um so there would be the I mean that comes from some measurement issue right so so when you measure it and you confuse say spider for flower or flower for spider then you would get some error and then maybe that's where I guess you can talk about the Marco blanket uh with specifying or misobserving something so I I'm not sure what what exactly is thinking yeah I mean they're compatible right so in this case there's no error but if there was error obviously there would be something wrong empty let me actually just correct this I believe be I mean I I didn't mean to say this is the visual error this is saying that they're unchanging the flowers so it's either they're either one or the other here parametrically it's possible to introduce visual error in the a matrix but I'm I'm just pointing to the the architecture of the model doesn't require even engaging with this it's totally compatible and the observations are the set States and the actions are the active States and so on but as presented and as constructed we if we're going to use a pomdp architecture we don't necessarily even need a representation like this yeah Jessica okay that's interesting [Music] hi yes I for me the reason why like it's like two part like the reason why they they send um crap is so popular and I think maybe it's also the reason why it was used and this is the Simplicity to show the concept and the relationship so like the new one it it's great for I guess for the paper that it was created but it doesn't show the relationship as clear as the other ones it's very

simple so I think that's why it was used maybe like we move a lot of extra information that the second paper illustrate like study in it but the Simplicity of that graph is why like it really encapsible the concept and it's also the reason why it's so popular because it's so simple so I think that's you know with a relationship I think I don't know how I see it yep totally makes sense thank you um and once we see that there is a mapping from the schematic cybernetic View then we can probably fill in a lot of the parts of the pmdp but just to look One More Level here I mean the code I posted the code below so it's in Matlab which people may or may not have access oh sorry okay um Ali and then Ron no I just wanted to point out one thing that I forgot to mention before uh one thing that might be helpful to keep in mind is that actually in this generative model there is a kind of unification or kind of integration of interceptive and extraceptive modalities because uh as they pointed out in the paper uh this agent this agent that they're modeling uh adopt two modes of Engagement with the world uh one of which is the relaxed and the other one is uh the arouse aroused mode of Engagement and because of that there are two sorts of hidden States related to each intercept of an extraceptive one uh so I guess uh if they wanted to put that kind of uh into relationships between uh these kind of modalities in that uh simplified schematic model it wouldn't necessarily clarify these kind of integration between the the modalities so one reason might be related to that I mean peculiar a case of a particular situation of this particular case here thanks it also speaks to the step in the model of the form of a generative model and then I think when people get married um the cybernetic scheme you can imagine well then there's two nodes inside of the internal States there's two modes you can go between but now you're like mixing and matching what arrows mean where some of the arrows are reflecting causal influence and some of them are reflecting transitions between mutually exclusive States and so then we end up back into the wireframe nightmare where it's like a schematic where some edges are tactile connections and some are causal links and some are mutually exclusive transfers of information and all these other ways that people visualize models and this representation reminds us that we're dealing with variables and states and their connections Jessica so I think that was from before okay but yeah but I guess yes I think it's like the visual it has to do with what are you trying to convey in that moment and you know how much depth of information do you need I think like at least say like for this one you need extra information so that all the model wouldn't work well just share like with the idea of what the model is maybe this version with the extra information might be too complex for the other ones there's a better purpose but it's like what is the purpose he's trying to you know accomplish and what do you want to communicate and

depending on that then models with more information and less information are more successful yep thank you totally agree I mean they didn't do any of the loop at all they just went straight to the pmdp representation and also this isn't the only form of an interoceptive extrareceptive integration model one could brainstorm a whole host and again that's what the recipe is for well we could have a mutually exclusive transfer between these two states defined by these two B matrices which entails a policy variable switching between the two or we could have a continuous variable called arousal and arousal is engaged in some sort of attenuation relationship so this isn't even the only form but it shows us a form soon in 20 minutes we're going to be heading into chapter 7. so let's preview it so that when we're working on this in the next week preparing our questions starting to think about what we want to be modeling we'll kind of know what's going to happen all right chapter seven is active inference in discrete time people can still hear me okay I think so all right here is where we crossed the Rubicon with modeling this is abstract up till now you cut out there for a second oh okay okay I'm it's all good now uh yeah just okay let's start from the review thank you chapter seven we're Crossing the Rubicon together and um it's been abstract up till now first what is addressed is perceptual processing you'll find a lot of resonance between this chapter 7. step by step which is model stream one maybe it's step hyphen yeah this paper step-by-step tutorial is going to do a certain walk through of adding incremental complexity into the model in terms of model parameter depth okay it's model stream one there's four streams that Ryan and Chris White did static perception Dynamic perception Dynamic perception with preferences and free energy minimization Dynamic perception with free energy minimization and flexible policy selection through uncertainty modulation we're going to do something similar in chapter 7. they don't do that upper left corner of Step by Step they go straight to Hidden Markov models for dynamic perception this is analogous to a common filter or any other number of General Bayesian filtering schemes we have a prior d hidden States s and a sequence of observables through a Time T that are mapped to Hidden States a b describes how hidden States change their time so we have the temperature is a hidden variable and then we're getting observations from the thermometer that are mapped to the real temperature by a Maybe it's a good thermometer maybe it's biased or whatever and then there's the temperature changing Through Time then there's a very um ali-centric discussion of listening to music this has to be one of the most glaring errors in the textbook it's very Warhol like though so maybe it has a certain aesthetic to it um but the example that's going to be provided is like a musical listening example um the a matrix is emitting notes based upon the hidden state which is the score of the music the B

Matrix is and seven they have 70 accuracy so this one-tenth distributes through the Matrix so it's just to use like integers here but really this is like 0.7.1.1.1 because it has to add up to one but when this person when the hidden state of the score is in the first note they play it seventy percent of the time and then the other three notes they make an error ten percent of the time each time um seventy percent of the time the musician hits their note the B Matrix is how things change through time the hidden state so this is the transitions in the score and here there's a 97 probability again it's distributed throughout with this one over a hundred just to represent it with integers and there's a prior with no uncertainty or ambiguity of just it begins in the first position and so this we can even see like you know I I don't want to use ABCD but it would be like b c d a in the musical ontology so here's b c d a like that but then there is an error in this one simulation run it'd be probabilistic but they're showing for didactic reasons a run where the third what point in the third observation there is an error however it's heard as the correct value that's a pure inference example this is not active inference this is just Bayesian models of perception this kind of sessile creature is rather uninteresting Now we move from a hidden Markov model Crouching Tiger Hidden Markov into a partially observable Markov decision process a hidden Markov model is a partially observable Markov model because some parts of it are hidden some are not but the key piece here is the introduction of the decision we know that we model that with policy that influences how States change Through Time and that policy selection is governed by G figure four three people can still hear because I see all these like Frozen for me okay okay all right okay thank you yeah it's just strange stuff um here's four three again when we're talking about policy selection we're talking about states that haven't happened yet observations and hidden states that haven't happened yet we see a restatement of equation 2.6 with expected free energy here broken into epistemic and pragmatic value now to flesh out this pomdp we turn to a classic par and friston which is the teamace we're not going to go through every single part of this teammate right now but it'll be a key area for us to unpack why do the matrices have the dimensions that they have where are the where are the four locations that the mouse can be you know here down at the bottom to the left or to the right so this is not like a mouse moving in continuous space it's like there are four locations that can be in those are the columns there's five hidden states of location which we can see are Ike based upon whether when it goes down to the bottom to get a q whether the hidden state is going to reveal the Q or not and then figure seven four and seven five you'll see differ first off one of them is bolded this one is bolded this one's unbolded but here we see a one in the L column because the black is on the L and here

we see a one in the r column because the block is on the r so understanding why that is is going to be the Crux of understanding how to construct an epistemic informative cue as a hidden environmental variable in a generative model here are the B matrices transition probabilities here's the preference vector we'll talk about what $C1$ and $C2$ are here are the priors we'll talk about what $D1$ and $D2$ are now there's more unpacking of the epistemic value itself so in the equation above there was a restatement of equation 2.6 which partitions the expected free energy into the negative epistemic value and the pragmatic value now we're going to follow up on the epistemic value I of P_i the epistemic value of a policy and break that down into these representations here are simulations simulated epistemic and pragmatic behavior of a rat foraging in a teammates so we'll try to model what these runs are I think it might be one small note about the book that especially if they knew that it was going to be black and white which it seems like that was known this is not Ultra helpful perhaps people disagree but another one this is what said another Warhol yeah this one this one you can even see the Campbell Soup but this I think if someone was familiar with reading these traces they would still want to understand what these different lines were and like what the dashed one is but if people are unfamiliar reading these kinds of error residual traces I don't know what is being shown here figure 7.7 that was the simulation of the Rat getting the epistemic cue and going to the left so instead of just taking a 50/50 shot on where the food was the rat navigates this explore/exploit dilemma it engages in an epistemically oriented action or an action driven by epistemic to to reduce uncertainty get the cue and then it secures the food in this box 7-1 we see a discussion about parameterization of uncertainty and how Precision is the inverse of variance and this Ω is the Precision variable here just like Mark Salms uses but others use different ones the lower this value is the more even things are and so this this has come up in the ice cream example with the folk psychology live stream like if your C Vector is 0.6 negative six versus zero 500 negative 500 that's like a sharper preference distribution so it's the quantitative values in the C for example that's specified not just the direction but also the intensity of preference and then analogously Precision can be specified in a continuous form here is a reference to a earlier par and Friston work on vision that has to do with ocular motor foraging which is heavily epistemically driven but then um are using this example to explore how precision the viewer shows a version to the upper left Square when it is specified with a less precise more ambiguous likelihood mapping the lower left square is epistemically attractive when the transition probabilities are specified as more uncertain we'll try to understand these examples learning a novelty 7.5 here we take that pomdp and like we push it

back in the genealogy we have a priors on the hidden states which are themselves equipped with prior beliefs these are dear shly conjugate priors but they can be of other forms but they're just shown as dirichlet here because it's like a convenient statistical distribution family for variational inference which we explored in several chapters ago um these can be considered as hyper priors because they they but in the end just variables on a graph and it's not like they're tagged with being a hyper prior and you can imagine priors that Nest on them exploring these potentially open-ended Bayesian graph structures is why it's so important to have structure learning like Carl has pointed out for years and why we want to have parameter sweeps across cognitive model structures in active block prints Because Why Stop why not have another prior on this one so we should have the notation and the capacity to sweep that here is a discussion on conjugate priors a conjugate prior means that when used to perform Bayesian inference the posterior belief will be the same type of distribution so previously we talked about how like the hidden State could be a continuous variable and then the internal State could be like a discrete variable you know the temperature is continuous outside and then inside is it hot or is it cold is like what I'm caring about so the families don't have to match but when we use conjugate priors it makes it so that where you can design it to play nice it does play nice but we'll work through that also with anybody who wants to go into more detail on the math and because these are just like Bayesian statistics points here Theta is used to just describe parameters it's a very common notation but not 100 pervasive to use Theta for parameters just generally here there's more unpacking of the update rules this is probably of secondary consideration for just constructing a model here we return to expected free energy and several more representations Beyond What was seen above and beyond what was seen in equation 2.6 here the learning model that was presented above the um pomdp with another generation of genealogy is described which is drawn from these two papers then structured learning is introduced we'll see how deep this one goes but these papers are quite interesting and connected to Bayesian model reduction here a hierarchical model is presented here's figure four three in the dashed box now there's a figure four three you know yo dog I heard you liked figure four three then there's another hierarchical model simulation from the above showing belief updating again with these Trace diagrams There's a summary so it's a somewhat long chapter it contains like the music example the rat example the maze example the eye example so we're seeing like data points in model space in the last three minutes does anyone have any thoughts or questions how are they going to tackle chapter seven and as they start to construct their model if they want to I am just a every direction click that I can because

it is quite dense still and kind of unapproachable I think for me anyways it just is quite difficult to like get a grasp on it from the textbook specifically at this point like the degenerative model like what does that mean in code and where does that correlate to this particular function or that particular time step or whatever um it's you know I don't know it's just a lot and so I'm just trying to from yeah I don't know from just generally building up simulations which are just things that you can you know put together in a browser like with simple JavaScript to like CAD Cad and active block rents and you know these sorts of things and trying to grasp at some real world situation so maybe that that rat you know teammates is a really good place to focus on um for me like a good intersection between the book and something that isn't like the frog jumping or is it not jumping is you know so contrived it's really difficult to think about modeling that and in a way that would help explain what's going it's like yes I knew the Frog was jumping or not jumping so that doesn't really help right um whereas the rat maybe coming up with some left or right hidden State based on the queue that's partially it's there's some explanatory power there that maybe realized I think so um I'm just yeah I don't know it's I'm trying to bridge jump over the Grand Canyon of the philosophical conceptual okay got it but the math there's a okay but there's a gap there for me so I'm trying to pull them together and it's uh yeah I have a question related to calibration of those priors and among other things uh is that ever explained as to how to go about doing that for any of these models model based data analysis is going to describe okay so it's it's okay it's in the future all right yeah I'll be for that chapter seven describes just in our last minute chapter 7 describes the discrete State models chapter 8 describes continuous time models chapter 9 describes using empirical data for model driven data analysis chapter 10 is a summary the textbook is you know yeah we're excited towards the Grand Canyon but I think again we have several months to unpack this together so I hope that we can go quite far with this so um please stay engaged and add your questions and start building from both sides play with the code examples provided break them and then start from the other side with a recipe on a situation that you care about and then they're going to meet in the middle thank you everyone farewell bye